DRAFT — NOT FOR PRODUCTION — 4 OPEN ITEMS

WORK INSTRUCTION

DESK LAMP ASSEMBLY

Number	WI-DL-001 Rev A
Assembly	desk_lamp.step
Part number	desk-lamp
Status	DRAFT

SCOPE

Putting one desk lamp together from its printed parts, its steel weight, the LED module and its screws.

APPLICABLE DOCUMENTS

None cited.

SAFETY

None stated.

PARTS

PART	QTY	STEP	OCCURRENCES
base	1	1	#o1.1
weight	1	2	#o1.2
head	1	3	#o1.5.4.1
led_module	1	4	#o1.5.4.2
m3x8_screw:left	1	5	#o1.5.4.3
m3x8_screw:right	1	5	#o1.5.4.4
arm	1	6	#o1.5.1
m5x30_screw:base_hinge	1	7	#o1.3
m5_locknut:base_hinge	1	8	#o1.4
m5x30_screw:head_hinge	1	10	#o1.5.2
m5_locknut:head_hinge	1	11	#o1.5.3

EQUIPMENT

ITEM
4 mm hex key
8 mm spanner
2.5 mm hex key

OPERATIONS

OP	OPERATION	STEPS
1	Weight the base	1–2
2	Fit the LED module into the head	3–5
3	Hang the arm on the base	6–8
4	Hang the head on the arm	9–11

DRAFT — NOT FOR PRODUCTION — 4 OPEN ITEMS

OP	OPERATION	STEPS
5	Set and check the hinges	12

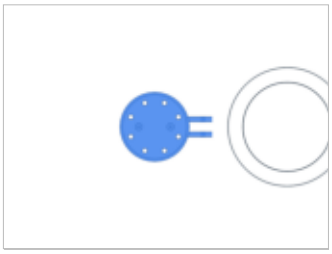
OPEN ITEMS

- step 2 (bond): adhesive not stated
- step 5 (fasten): torque not stated
- step 8 (fasten): torque not stated
- step 11 (fasten): torque not stated

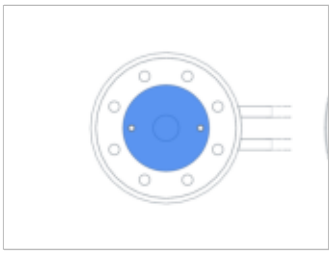
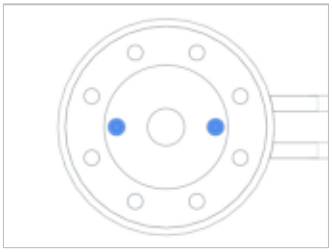
DRAFT — NOT FOR PRODUCTION — 4 OPEN ITEMS

PROCEDURE

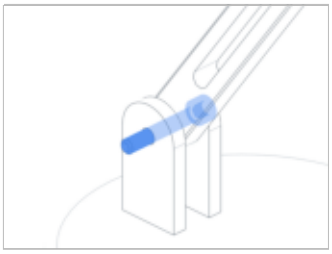
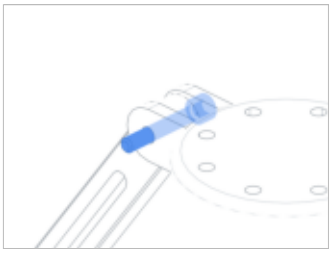
FIGURES: the assembly after the step, in line art. Filled blue: the parts this step installs. Purple: the surfaces it names. Outline only: parts installed earlier.

STEP	INSTRUCTION	FIGURE	DONE
OPERATION 1	**Weight the base**		
1	**Lay the base upside down on the bench** Install: base (#o1.1)		
2	**Bond the steel weight into the base's pocket, flush with the underside** Install: weight (#o1.2) Surfaces: base (#o1.1.f15) **Adhesive:** NOT STATED Inspect: The weight sits flat on the pocket's ceiling, its underside flush with the base's FLUSH, YES OR NO _____		
OPERATION 2	**Fit the LED module into the head**		
3	**Lay the head on the bench, its opening up** Install: head (#o1.5.4.1)		

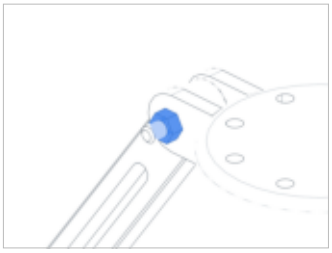
NOTE
Lead the LED's wires up through a vent beside the yoke before the board goes on.

4	**Set the LED module on the head's two bosses, its emitter out of the shade** Install: led_module (#o1.5.4.2)		
5	**Screw the LED module to the bosses** Install: m3x8_screw:left (#o1.5.4.3), m3x8_screw:right (#o1.5.4.4) Hardware: ISO 4762 M3 x 8 socket head screw, qty 2 **Torque:** NOT STATED The screws cut their own thread in the bosses' 2.5 mm pilot holes: drive them until the board is held, no further.		

DRAFT — NOT FOR PRODUCTION — 4 OPEN ITEMS

STEP	INSTRUCTION	FIGURE	DONE
OPERATION 3	**Hang the arm on the base**		
6	**Turn the base over and put the arm's lower end between the clevis's cheeks** Install: arm (#o1.5.1)		
7	**Push the hinge screw through the clevis and the arm, its head on the cheek nearest you** Install: m5x30_screw:base_hinge (#o1.3)		
8	**Run the locknut onto the hinge screw** Install: m5_locknut:base_hinge (#o1.4) Hardware: M5 nylon insert locknut on the M5 x 30 hinge screw, qty 1 **Torque:** NOT STATED		
OPERATION 4	**Hang the head on the arm**		
9	**Put the head's yoke over the arm's upper end, the shade in front**		
10	**Push the hinge screw through the yoke and the arm** Install: m5x30_screw:head_hinge (#o1.5.2)		

DRAFT — NOT FOR PRODUCTION — 4 OPEN ITEMS

STEP	INSTRUCTION	FIGURE	DONE
11	**Run the locknut onto the hinge screw** Install: m5_locknut:head_hinge (#o1.5.3) Hardware: M5 nylon insert locknut on the M5 x 30 hinge screw, qty 1 **Torque:** NOT STATED		
OPERATION 5 Set and check the hinges			
12	**Tighten each hinge's locknut until the arm and the head hold wherever they are set** Inspect: With the arm at its lowest, 10 degrees up from the desk, it holds there and the lamp stands HOLDS, YES OR NO _____ Inspect: With the head square to the arm as assembled, the arm swings from 10 degrees up from the desk, over the top, to 10 degrees below level behind, touching nothing CLEAR, YES OR NO _____ Inspect: With the arm at 55 degrees up, the head swings from 45 degrees down to 175 degrees up from hanging level, touching nothing CLEAR, YES OR NO _____		